

Santiago Herrera Tafur

Junior Data Engineer | Systems Engineer | Master's in Big Data

Lisbon, Portugal (on-site, hybrid or remote) | +351 920 787 540 | santyherrera12@gmail.com

LinkedIn: [linkedin.com/in/santiago-herrera-tafur-463887161](https://www.linkedin.com/in/santiago-herrera-tafur-463887161) | GitHub: github.com/santyherrera12 | Portfolio:

santyherrera12.github.io/Portafolio

PROFESSIONAL SUMMARY

Systems Engineer completing a Master's in Big Data at Universidad Europea de Madrid, focused on Data Engineering. Portfolio of 14 completed projects covering ETL pipelines, Medallion architecture (Bronze/Silver/Gold) with dbt and PostgreSQL, real-time streaming with Apache Kafka, distributed processing with Apache Spark (PySpark), REST APIs with FastAPI and containerized deployments with Docker. Solid SQL (JOINs, CTEs, window functions) and Python (Pandas). Seeking a Junior Data Engineer, Data Analyst or Backend Developer role; open to other IT areas.

TECHNICAL SKILLS

Programming: Python, SQL, Shell scripting (Bash), JavaScript, HTML, CSS

Data Engineering: ETL pipelines, dbt, Apache Spark (PySpark), Apache Kafka, Data Lake, Data Warehouse, Medallion architecture (Bronze/Silver/Gold), web scraping (BeautifulSoup)

Databases: PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, Google BigQuery

Cloud and DevOps: Docker, Docker Compose, Git, GitHub, Linux/Unix, AWS (S3, IAM)

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, Swagger UI

Data Analysis and ML: Pandas, Scikit-learn, Random Forest, Prophet, time series forecasting, anomaly detection

Data Visualization: Power BI, Streamlit, Plotly, Matplotlib, React (Recharts)

Data Governance: Data catalog, business glossary, data lineage, data quality rules, GDPR

Tools: VS Code, Google Colab, Microsoft 365 (advanced Excel)

PROJECTS

All code available at github.com/santyherrera12

Petroleum Trade Balance Analysis and Forecasting Platform — Master's Thesis (in progress)

Python, Pandas, Prophet, Streamlit, Plotly, Data Lake

- Built a Bronze/Silver/Gold Data Lake combining CNMC import and export data (3,906 records, 14 products, 2003–2026) through a modular pipeline of five Python modules (bronze, silver, gold, forecast, visualize).
- Forecasted Spain's petroleum balance to 2030 with Prophet; implemented automatic anomaly detection and impact analysis of the 2008 crisis, COVID-19 and the 2022 energy crisis.
- The analysis shows Spain reduced its net petroleum dependency by 71% since 2003; results presented in an interactive Streamlit app.

Data Warehouse with dbt and PostgreSQL

dbt, SQL, PostgreSQL, Docker

- Medallion Data Warehouse (Bronze, Silver, Gold) with 9 version-controlled SQL models and 17 automated data quality tests.

Real-Time Streaming Pipeline

Python, Apache Kafka, MongoDB, Docker Compose

- Containerized pipeline where a producer generates a transaction every 2 seconds, publishes it to Apache Kafka, and a consumer persists it in MongoDB.

Apache Spark Data Pipeline

PySpark, Python

- Distributed processing of 100,000 transactions: data cleaning, distributed aggregations and window functions (RANK, LAG).

REST API with FastAPI and PostgreSQL

Python, FastAPI, SQLAlchemy, Pydantic, PostgreSQL, Docker

- Layered backend with CRUD endpoints, Pydantic validation, SQLAlchemy ORM, PostgreSQL running in Docker and Swagger UI documentation.

Advanced SQL Analytics

SQL, PostgreSQL, Docker

- Business queries on a 5-table e-commerce database using multiple JOINs, subqueries, CTEs and window functions (RANK, LAG, PARTITION BY): revenue per customer, most profitable months and average order value trend.

Machine Learning Pipeline — Employee Attrition Prediction

Python, Scikit-learn, Pandas, FastAPI

- Random Forest model (95.5% accuracy) with training, evaluation metrics and a real-time prediction API built with FastAPI.

Cloud Analytics Dashboard with BigQuery and Power BI

Google BigQuery, SQL, Power BI

- Analytical SQL queries on a video game sales dataset in BigQuery answering business questions; executive dashboard with 5 visualizations connected directly to BigQuery.

ETL Pipeline — Weather Data

Python, Pandas, REST API, python-dotenv

- Modular Extract, Transform and Load pipeline pulling weather data for multiple cities from the OpenWeatherMap API into a historical CSV; virtual environment and .env configuration.

Web Scraping and Automated Analysis

Python, BeautifulSoup, Pandas

- Extracted 980 books from 50 web pages, cleaned the data with Pandas and generated an analysis report (scraper, analysis and main modules).

Data Governance — Airbnb Madrid Dataset (academic)

Python, Google Colab

- Business glossary, data catalog, data lineage and GDPR analysis; data quality rules implemented in Python.

Other projects: Full stack app (React, FastAPI, PostgreSQL, Docker); interactive sales analytics app (Streamlit, Plotly); analytics dashboard (React, Recharts); JavaScript web app; responsive landing page (HTML, CSS); D3.js data visualization.

EDUCATION

Piscine — Intensive Selection Program | Sep 2026

42 Lisboa (42 Network) — Lisbon, Portugal

- Completed the 26-day full-time Piscine: daily programming exercises in a Unix/Linux environment, shell scripting (Bash) and version control with Git, under 42's peer-to-peer learning methodology.

Master's Degree in Big Data | Nov 2025 – Sep 2026 (expected)

Universidad Europea de Madrid — Madrid, Spain (Online)

- Modules: NoSQL databases (MongoDB, Cassandra, Redis), Apache Kafka, Hadoop, Apache Spark (PySpark), cloud computing architectures (AWS, Docker, Kubernetes), Google BigQuery, machine learning (Random Forest, K-Means, neural networks), data visualization (Power BI, D3.js), data governance, Big Data project management, Scala.

Bachelor's Degree in Systems Engineering | Aug 2019 – Oct 2025

Universidad Cooperativa de Colombia — Ibagué, Colombia

- Software development, relational databases (SQL, Oracle), database design and administration, networking and communications, systems architecture.

Systems Technology Degree | Jan 2016 – Jun 2017

SENA — Centro de Comercio y Servicio, Colombia

- Information systems analysis and development.

CERTIFICATIONS AND COURSES

- Oracle Database SQL (1Z0-071) — in preparation
- Web Design and Development — SENA (2019)

LANGUAGES

Spanish: Native | English: B1-B2 (currently taking classes) | Portuguese: B1 (currently taking classes)

SOFT SKILLS

Problem solving, analytical thinking, self-directed learning, teamwork in technical environments, continuous improvement.